

Systems of Linear Equations — 30 MCQs

Digital SAT Style | Easy / Medium / Hard

Q1. DIFFICULTY: EASY

$$3x + 2y = 14$$

$$4x - 2y = 7$$

If (x, y) is the solution to the system of equations above, what is the value of x ?

- A) 2
 - B) 3
 - C) 4
 - D) 5
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Q2. DIFFICULTY: EASY

A summer camp sells two types of tickets: standard tickets for \$12 each and premium tickets for \$20 each. On Monday, the camp sold a total of 35 tickets and collected \$580. Which of the following systems of equations can be used to find the number of standard tickets, s , and premium tickets, p , sold that day?

- A) $s + p = 35$ and $12s + 20p = 580$
 - B) $s + p = 580$ and $12s + 20p = 35$
 - C) $s + p = 35$ and $20s + 12p = 580$
 - D) $12s + 20p = 35$ and $s + p = 580$
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Q3. DIFFICULTY: EASY

$$5x - y = 11$$

$$x + y = 7$$

What is the value of y in the solution to the system above?

- A) 1
 - B) 2
 - C) 3
 - D) 4
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Q4. DIFFICULTY: EASY

Line A passes through the points $(0, 5)$ and $(5, 0)$. Line B has the equation $y = 2x - 4$. What is the x -coordinate of the point where Line A and Line B intersect?

- A) 2
- B) 3
- C) 4

D) 5

Q5. DIFFICULTY: EASY

$$4x + 3y = 24$$

$$4x - y = 8$$

What is the value of y ?

A) 2

B) 4

C) 5

D) 6

Q6. DIFFICULTY: EASY

Two friends are saving money. Layla has \$45 and saves \$15 per week. Marcus has \$10 and saves \$25 per week. After how many weeks will they have the same amount of money?

A) 2.5

B) 3

C) 3.5

D) 4

Q7. DIFFICULTY: EASY

$$y = 3x - 2$$

$$y = -x + 6$$

Which of the following is the solution (x, y) to the system above?

A) (1, 1)

B) (2, 4)

C) (3, 3)

D) (4, 2)

Q8. DIFFICULTY: EASY

If $2x + y = 9$ and $x - y = 3$, what is the value of $x + y$?

A) 5

B) 6

C) 7

D) 8

Q9. DIFFICULTY: EASY

A taxi service charges a flat fee of \$3.50 plus \$1.75 per mile driven. A rideshare service charges a flat fee of \$5.00 plus \$1.25 per mile. For what number of miles will the two services charge the same total amount?

- A) 2
- B) 3
- C) 4
- D) 5

Q10. DIFFICULTY: EASY

$$6x + 2y = 20$$

$$3x + 2y = 14$$

What is the value of x ?

- A) 1
- B) 2
- C) 3
- D) 4

Q11. DIFFICULTY: MEDIUM

$$3x + ky = 12$$

$$6x + 8y = 24$$

In the system of equations above, k is a constant. If the system has infinitely many solutions, what is the value of k ?

- A) 2
- B) 4
- C) 6
- D) 8

Q12. DIFFICULTY: MEDIUM

A bakery produces muffins and croissants. Each muffin requires 0.5 cups of flour and each croissant requires 1.5 cups of flour. Each muffin requires 1 egg and each croissant requires 2 eggs. The bakery uses exactly 15 cups of flour and 22 eggs each morning. How many croissants does the bakery make each morning?

- A) 4
- B) 6
- C) 8
- D) 10

Q13. DIFFICULTY: MEDIUM

$$2x - 5y = 3$$

$$4x + py = 9$$

For what value of p will the system of equations above have no solution?

- A) -10
 - B) -5
 - C) 5
 - D) 10
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Q14. DIFFICULTY: MEDIUM

The sum of two numbers is 54 and their difference is 18. What is the product of the two numbers?

GRID-IN

[GRID-IN — enter your answer in the box]

Q15. DIFFICULTY: MEDIUM

$$ax + 4y = 16$$

$$3x + 12y = c$$

In the system above, a and c are constants. If the system has infinitely many solutions, what is the value of $a + c$?

- A) 4
 - B) 16
 - C) 49
 - D) 64
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Q16. DIFFICULTY: MEDIUM

A store sells two types of memberships: a basic plan for \$8 per month and a premium plan for \$15 per month. In January, the store sold 200 memberships for a total monthly revenue of \$2,020. How many premium memberships were sold?

- A) 60
 - B) 80
 - C) 100
 - D) 140
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Q17. DIFFICULTY: MEDIUM

If $3a - 2b = 11$ and $a + 2b = 9$, what is the value of $2a$?

- A) 10
- B) 12
- C) 14
- D) 16

Q18. DIFFICULTY: MEDIUM

Two candles are lit at the same time. Candle A is 24 centimeters tall and burns at a rate of 2 centimeters per hour. Candle B is 16 centimeters tall and burns at a rate of 1 centimeter per hour. How many hours after they are lit will the two candles be the same height?

- A) 4
- B) 6
- C) 8
- D) 10

Q19. DIFFICULTY: MEDIUM

$$mx + 4y = 12$$

$$5x + ny = 30$$

If the system above has infinitely many solutions, and m and n are constants, what is the value of m/n ?

- A) $1/5$
- B) $2/5$
- C) $5/2$
- D) 5

Q20. DIFFICULTY: MEDIUM

Two gym packages are available. Package A costs \$50 to join and \$30 per month. Package B costs \$10 to join and \$40 per month. After how many months will the total cost of both packages be equal?

GRID-IN

[GRID-IN — enter your answer in the box]

Q21. DIFFICULTY: MEDIUM

The graphs of $y = 3x + 7$ and $y = kx - 5$ are parallel lines in the xy -plane. Which of the following is the value of k ?

- A) -3
- B) $-1/3$
- C) $1/3$
- D) 3

Q22. DIFFICULTY: MEDIUM

A parking lot charges \$4 for the first hour and \$2 for each additional hour. A nearby garage charges \$6 for the first hour and \$1.50 for each additional hour. Let h represent the number of

additional hours (after the first) parked. Which value of h results in the two places charging the same amount?

- A) 2
 - B) 3
 - C) 4
 - D) 5
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Q23. DIFFICULTY: HARD

$$(k - 2)x + 3y = 9$$

$$2x + (k + 1)y = 12$$

For what value of k will the system of equations above have exactly one solution where $x = 0$?

- A) 1
 - B) 2
 - C) 3
 - D) 5
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Q24. DIFFICULTY: HARD

A scientist mixes two solutions. Solution X contains 20% acid and solution Y contains 50% acid. She wants to make 60 liters of a solution that is 30% acid. How many liters of Solution Y should she use?

GRID-IN

[GRID-IN — enter your answer in the box]

Q25. DIFFICULTY: HARD

$$6x - 9y = 15$$

$$2x - 3y = c$$

The system above has no solution. Which of the following CANNOT be the value of c ?

- A) 3
 - B) 4
 - C) 5
 - D) 7
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Q26. DIFFICULTY: HARD

The system below has no solution:

$$5x - 3y = 7$$

$$10x + cy = 20$$

What is the value of c ?

- A) -6
 - B) -3
 - C) 3
 - D) 6
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Q27. DIFFICULTY: HARD

In the system of equations below, a and b are constants.

$$(a + 1)x + 3y = b$$

$$4x + 6y = 12$$

If the system has infinitely many solutions, what is the value of ab ?

- A) 6
 - B) 12
 - C) 18
 - D) 24
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Q28. DIFFICULTY: HARD

A concert venue sells floor seats and balcony seats. On night 1, they sold 120 floor seats and 80 balcony seats for \$8,800 total. On night 2, they sold 90 floor seats and 150 balcony seats for \$9,750 total. What is the price of one balcony seat?

GRID-IN

[GRID-IN — enter your answer in the box]

Q29. DIFFICULTY: HARD

$$y = 2x^2 - 8x + 6$$

$$y = 2x - 2$$

How many solutions does the system above have?

- A) 0
 - B) 1
 - C) 2
 - D) Infinitely many
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Q30. DIFFICULTY: HARD

The population of Town A is modeled by $P = 4,000 + 300t$ and the population of Town B is modeled by $P = 6,500 - 200t$, where t is the number of years since 2020. In what year will the two towns have equal populations?

- A) 2024
- B) 2025

- C) 2026
 - D) 2027
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